Prabhat Kumar

B.S. Mathematics and Computing
Indian Institute of Technology, Kharagpur

+91-7367905043
sauravsandilya7367905043@gmail.com
linkedin.com/in/prabhat-kumar-654b7b413
github.com/Prabhat-190

Education

Indian Institute of Technology, Kharagpur **CGPA: 7.29**
Bachelor of Science in Mathematics and Computing

Experience

Software Engineering Intern *May 2026 – July 2026*
Paycraft github.com/Prabhat-190/pbf-generator

- Built a **Java 21 / Spring Boot 3** settlement service that generates ABT-to-NLP **PBF** files in Header–Record–Trailer format, validates transit transactions, and tracks file lifecycle from creation through loyalty callback completion.
- Implemented **REST APIs** for PBF generation, status lookup, callbacks, and health checks; persisted file metadata, per-record results, and four-digit sequences using **Spring Data JPA** and **H2**.
- Engineered the file pipeline: business-rule validation, trailer reconciliation, **SHA-256** checksums, ZIP packaging, mock **SFTP** delivery, year/month archival, and a scheduled retry job for failed uploads.
- Added **SLF4J/Logback** logging with request-id tracing, global exception handling, **OpenAPI/Swagger** docs, **JUnit 5** unit and integration tests, and Dockerized the service for Railway deployment.

Projects

B.Tech Project: Machine Learning based Stock Prediction and Portfolio Optimization *Ongoing*
Python, Pandas, NumPy, Scikit-learn, Keras, Time-Series, Matplotlib

- Building a two-stage pipeline that forecasts next-day equity returns from **NSE/Nifty 50** OHLCV data and technical indicators, then allocates capital with **mean-variance (Markowitz)** optimization.
- Training and comparing Linear Regression, Random Forest, and **LSTM** models under walk-forward validation to avoid look-ahead bias; evaluating forecasts with RMSE/MAPE and portfolios with Sharpe ratio and maximum drawdown.
- Engineering SMA, EMA, RSI, MACD, and volatility features; imposing long-only and budget constraints and benchmarking optimized weights against equal-weight and index buy-and-hold strategies.

Algo Trade Simulator Live Project
Python, Dash, Plotly, Redis, WebSockets, Docker, Gunicorn, Nginx

- Developed an **algorithmic trading simulator** ingesting Level-2 order-book streams over WebSockets to estimate execution cost across slippage, market impact, fees, liquidity, and volatility.
- Designed a **4-layer Redis Pub/Sub** pipeline for market-data ingestion, order-book processing, cost analysis, and live visualization.
- Deployed with **Docker, Gunicorn, and Nginx** (multi-worker processing, WebSocket routing, rate limiting, Redis failover) and **100+ tests** covering simulation, feeds, and persistence.

Amazon FinAI: Smart Checkout and Conversational RAG Engine GitHub
React, Node.js, FastAPI, MongoDB, Redis, ChromaDB, OpenAI, Docker

- Built a **fintech microservice** platform combining payment routing and generative AI for checkout, budget tracking, and assistance.
- Implemented a **RAG** engine (GPT-4, ChromaDB, tool calling) and ranked payment methods on **3 Redis signals**: gateway success, cashback, and gift-card expiry.
- Extracted structured discounts from bank offers via **GPT-4 Vision** and containerized **3 services** (React, Node.js, FastAPI) with Docker Compose.

Fraud Shield AI Live Project
Python, Scikit-learn, Pandas, NumPy, Streamlit, CCXT

- Built an end-to-end **fraud detection** system using Random Forest to score transactions on a **0–100%** risk scale, with precision-recall threshold tuning to cut false-positive alerts.
- Engineered amount, time, cardholder age, geography, and merchant-category features; deployed a **Streamlit** dashboard with live scoring and **CCXT** market-data fallback.

Technical Skills

Languages:	Java, Python, C++, JavaScript, SQL, React.js, Next.js, HTML, CSS
Backend:	Spring Boot, Spring Data JPA, REST APIs, Node.js, Express.js, FastAPI, WebSocket, Microservices, JUnit, SLF4J
ML / Data:	Scikit-learn, Keras, Pandas, NumPy, LSTM, Time-Series Forecasting, Portfolio Optimization, Plotly, Dash, Streamlit
Databases / Tools:	MySQL, MongoDB, Redis, H2, ChromaDB, Docker, Git, GitHub, Postman, Gradle, Gunicorn, Nginx, OpenAPI
Core CS:	Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design
Coursework:	Algorithms, DBMS, File Organization, Computer Organization, Theory of Computation, Big Data Analysis, Optimization, Probability and Statistics, Stochastic Processes, Linear Algebra, Discrete Mathematics, Graph Theory

Achievements

- **Codeforces Expert** (max rating **1736**); **Global Rank 245** among 25,673 participants in Round 1093 (Div. 2).
- Solved **500+** DSA problems across LeetCode, Codeforces, CodeChef, and HackerRank.
- Awarded the **Reliance Foundation Undergraduate Scholarship** (top **0.8%** of 120,000+ applicants).
- Secured **Top 8% in JEE Advanced** and **Top 1.2% in JEE Main**; HackerRank Software Engineer Intern Certification.